

Alexander Wegener

Robotics engineer — real-time control, teleoperation, and imitation learning.

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MSc Robotics, Boston University · graduating December 2026, available January 2027

German citizen · three years of US work eligibility via OPT and the STEM extension

EXPERIENCE

2026 —
present

Graduate Researcher, Robotics

MBZUAI · Abu Dhabi, operated from Boston

- Built a **full-stack teleoperation system operating Boston to Abu Dhabi** over the public internet — self-developed low-latency H.264/RTP video with FEC and adaptive quality control, plus per-device UDP state streams and a reliable acknowledged channel for commands that must not be dropped.
- Bimanual VR teleoperation: **Cartesian impedance control with nullspace posture regulation**, resolved-rate IK with damped least squares, pan-tilt joint impedance, and a **control-barrier-function self-collision filter** — running interchangeably against MuJoCo and libfranka.
- Unreal Engine 5** VR interface driven entirely by gaze and speech with an attention-adaptive layout, keeping both hands on the arms, plus a live HUD of measured end-to-end latency, jitter and loss.
- Collected and labelled a bimanual parcel-sorting dataset for intent recognition and imitation learning (**ACT**).
- Research question: does guiding an operator during collection produce demonstrations that train better policies? Paired policy comparison in progress.

04/2024 —
2025

Researcher — Teleoperation and Shared Control

MIRMI, TU Munich · Haddadin Lab

- Built a **distributed real-time robotic control stack** over a local network: 7-DoF robot arm, pan-tilt head, and passivity-based haptic feedback control.
- Ran a **ten-participant, six-condition, ~300-trial user study** on operator guidance — assistance cut NASA-TLX workload by over 40 points and median contact force from ~21 N to ~12 N with no cost in completion time.
- Co-authored "Enhanced Robotic Navigation in Deformable Environments using Learning from Demonstration and Dynamic Modulation," **IEEE/RSJ IROS 2025**.

09/2022 —
04/2023

Intern, Motor Controls

Tesla · System Architecture, Modeling & Controls

- Automated regression test tooling validating control algorithms across multiple thermal motor models; Python post-processing for test data analysis.

2018 —
2022

Technical Director, then Head of Drivetrain Development

Formula Student (Formula SAE) · Munich University of Applied Sciences

- Technical director for ~60 people across seven subteams** (two years): vehicle concept, cross-subsystem integration, budget, and sponsors including BMW, Knorr-Bremse and Infineon.
- Designed a **20,000 rpm, 29 Nm traction motor**, wrote its field-oriented control, and built the test bench that characterised it.

EDUCATION

2026

MSc Robotics — Boston University. *Expected December 2026.*

2023 — 25

MSc Controls & Automation — Technical University of Munich. Thesis at MIRMI, Haddadin Lab on operator nudging and shared control in teleoperation.

2018 — 22

BSc Electrical Engineering, electromobility focus — Munich University of Applied Sciences.

SELECTED PROJECTS

POLICY

Force-domain diffusion policy for sub-millimetre insertion. Contact-rich insertion from force alone, no vision. DDIM-10 reached 78% against the scripted expert's 61%, and transferred zero-shot to unseen peg geometries.

CONTROL

Vision-based NMPC for pushing under perception uncertainty. ArUco/PnP at 60 Hz feeding an EKF and acados SQP-RTI NMPC at 100 Hz; nominal, certainty-equivalent and chance-constrained control compared over a 360-run sweep.

SKILLS

CONTROL

Nullspace redundancy resolution, resolved-rate IK, control barrier functions, NMPC, LQR / iLQR, EKF

LEARNING

Imitation learning (ACT), diffusion policies (DDPM / DDIM), causal sequence models (GRU, transformer), reinforcement learning

SOFTWARE

C++17 (real-time), Python, MATLAB / Simulink · MuJoCo, libfranka, Pinocchio, acados, CasADi, PyTorch, Unreal Engine 5, OpenXR, GStreamer, CMake, Linux, Git · German (native), English (fluent)